AlkCalc: Alkali-metal atom and alkaline-earth-metal ion calculator

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Summary

AlkCalc (Euchner, 2026) is software designed to numerically compute properties of alkali-metal atoms and alkaline-earth-metal ions. More specifically, AlkCalc is designed to compute bound single-electron eigenenergies and eigenstates of the valence electron together with derived quantities such as transition matrix elements, oscillator strengths, and lifetimes. The eigenenergies and eigenstates are computed from the effective, single-electron model potentials given in (Aymar et al., 1996; Marinescu et al., 1994). The eigenstates and the derived quantities are of fundamental relevance in atomic, molecular, and optical (AMO) physics, but are also central to the design and characterisation of new technologies based on trapped atoms and ions. Examples include protocols for quantum information processing (Saffman et al., 2010; Sørensen & Mølmer, 1999, 2000; Zhang et al., 2020) and quantum simulations based on the enhanced properties of highly excited electronic states, so-called Rydberg states (Müller et al., 2008; Weimer et al., 2010).

AlkCalc is lightweight and portable software. The library itself is written in C and conforms to the C99 standard (ISO/IEC, 1999). The provided Makefiles are written for GNU make, but can be easily adapted to any other build system. AlkCalc further builds on decades-long tested FORTRAN routines adapted from Netlib (Dongarra & Grosse, 1987). However, this third-party software is built into AlkCalc directly and therefore does not constitute a dependency. Because of this design choice, it is straightforward to port AlkCalc to virtually any computing platform. Its core's only dependencies are a C library, a C compiler, and a FORTRAN compiler.

AlkCalc is thoroughly documented in the `README`s and `theory/theory.pdf`, which constitute its user's guide. The `README`s contain technical information on how to install and use AlkCalc. On the other hand, `theory/theory.pdf` provides comprehensive documentation of the library as well as a detailed introduction to the problems AlkCalc is designed to address and solve. It allows users who are new to the atomic and Rydberg physics communities to find all relevant information for the use of AlkCalc in a single, self-contained document. In AMO physics there can be different conventions for defining common quantities. One example is the Condon-Shortley phase for the spherical harmonics (Friedrich, 2017). AlkCalc explicitly documents all conventions used, eliminating possible ambiguities.

State of the field

For a long time, the Rydberg atom and Rydberg ion communities relied on small, independently developed, unpublished codes for numerically computing properties of low-lying electronic and Rydberg states. For Rydberg atoms, in particular the alkali metals, two community-driven open-source software packages unified these private codes: ARC (Šibalić et al., 2017) and PairInteraction (Weber et al., 2017). ARC is designed to compute properties of Rydberg states,

while PairInteraction focuses on the computation of interaction potentials between two Rydberg atoms. Both of these software packages extract the valence electron's eigenenergies from experimentally measured quantum defects that must be externally supplied. The corresponding radial wave functions are subsequently obtained by numerically integrating the radial Schrödinger equation at these eigenenergies using an effective, single-electron model potential to describe the valence electron, introduced in (Marinescu et al., 1994). Alternatively, PairInteraction offers analytical Whittaker functions to describe the eigenstates. A recent development in the numerical methods of the field is the treatment of divalent atoms, i.e., atoms with two valence electrons: with version 3.0, ARC was extended to the divalent atoms Ca, Sr, and Yb (Robertson et al., 2021). Recently, PairInteraction was likewise extended to divalent atoms using multichannel quantum defect theory, with states and matrix elements computed by the separate Python package RydState (Mögerle et al., 2026).

In contrast to alkali-metal atoms, there is no unified codebase for the electronic states of the valence electron in singly charged alkaline-earth-metal ions. Moreover, the available software packages ARC and PairInteraction are tailored to directly output derived electronic quantities, such as transition matrix elements, oscillator strengths, and lifetimes. These software packages are not designed to output and handle large subsets of eigenstates and their associated eigenenergies. In particular, these software packages do not include functionality to compute the eigenenergies directly from the model potential: the eigenenergies are externally provided by experimentally measured quantum defects. Up to this point there is no open-source software available that obtains *both* eigenenergies *and* eigenstates for alkali-metal atoms and alkaline-earth-metal ions directly from a single effective model potential. AlkCalc computes the eigenenergies and eigenstates directly from the model potentials given in (Aymar et al., 1996; Marinescu et al., 1994). Because this approach is different from ARC, PairInteraction, and RydState, which are all based on quantum defect theory, AlkCalc is implemented as a stand-alone open-source software project and not as an extension to the existing libraries.

Statement of need

The target audiences of the software AlkCalc are the atomic and Rydberg physics communities, including Rydberg-atom/-ion based quantum computing and simulation, metrology with trapped ions, sensing with Rydberg atoms, and atomic physics with energetically low-lying states, relevant for atomic clock experiments and optical lattices. AlkCalc addresses three key requirements of these communities:

(i) **Unified framework for both atoms and ions.** AlkCalc offers a unified codebase for the systematic treatment of Rydberg states in both neutral alkali-metal atoms and singly charged alkaline-earth-metal ions. To date, no open-source software package appears to be available for the latter.

(ii) **Efficient computation of eigenenergies and eigenstates.** AlkCalc computes *both* eigenenergies *and* eigenstates directly from the model potentials for alkali-metal atoms and alkaline-earth-metal ions given in (Aymar et al., 1996; Marinescu et al., 1994). This functionality has two benefits: first, in general, effective model potentials depend on parameters that must be numerically optimised such that the model potential reproduces the measured spectroscopic energies correctly. For fitting such parameters, it is therefore crucial to have a tool that allows one to efficiently compute the eigenenergies associated with a model potential. Secondly, for the theory side of the target communities, it is important to have software that allows one to efficiently compute eigenenergies and eigenstates. This is because in quantum physics this means the problem is solved: from the eigenenergies and eigenstates the full dynamics can be inferred and simulated on arbitrary time scales.

(iii) **Transparency and portability.** AlkCalc is self-contained and fully documented. Every step, from the model Hamiltonian to the stored eigenstates, is thoroughly described in

`theory/theory.pdf`, including all applied conventions in the formulation of the problems AlkCalc is designed to address. The code remains portable and usable in the long term, because AlkCalc does not build on interpreted programming languages. Its full functionality can be operated within C. This makes AlkCalc both a reliable foundation for future software in the steadily evolving Rydberg and atomic physics communities and a self-contained entry point for newcomers to the field.

Software design

AlkCalc follows a two-stage software architecture, which is visualised in Figure 1.

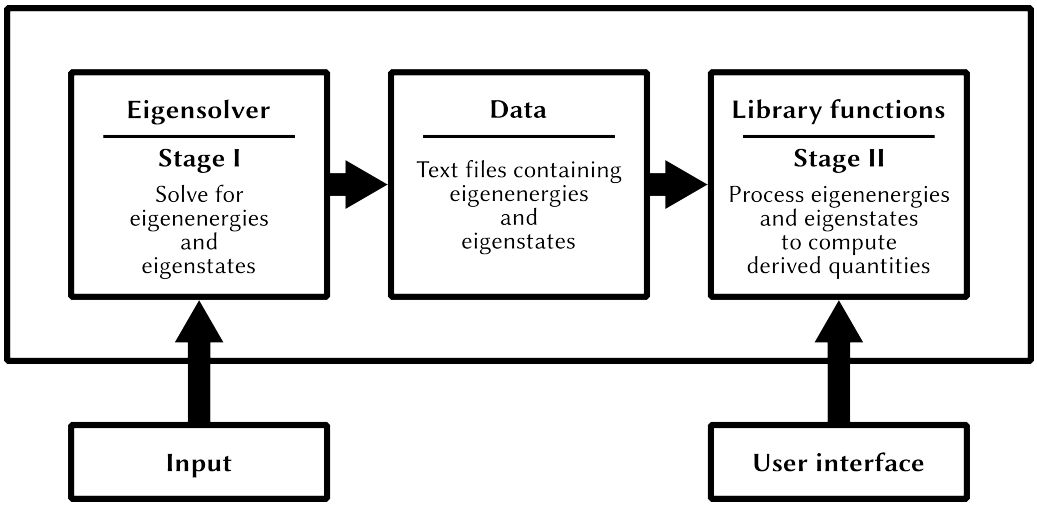


Figure 1: Software architecture. Visualisation of AlkCalc's software architecture. The leftmost box represents the eigensolver (Stage I), which is operated by the user indicated by the box labelled 'Input'. The computed eigenenergies and eigenstates are represented by the box in the centre. The rightmost box represents the library functions (Stage II). Through the user interface, represented by the lower right box, the library functions are called.

Stage I. This stage consists of an eigensolver that is tailored to solve for the eigenenergies and eigenstates of alkali-metal atoms and alkaline-earth-metal ions, described by a spherically symmetric, single-electron model potential in combination with the spin-orbit interaction contribution. Due to symmetry, the eigenstates factorise into an angular and a radial part. The angular part is solved analytically, as discussed in detail in the documentation of AlkCalc. The radial part is the solution to a one-dimensional boundary-value problem. AlkCalc employs B-splines of user-controllable order to approximate the radial eigenvalue problem with a banded, real-symmetric, generalised matrix eigenvalue problem. The relevant matrix elements are computed with de Boor's algorithm (de Boor, 2001) in combination with Gaußian quadrature (Golub & Welsch, 1969) for computing the associated integrals. The generalised matrix eigenvalue problem is subsequently solved with the LAPACK routine `DSBGVX` (Anderson et al., 1999). For evaluating matrix-vector products efficiently, the BLAS routine `DSBMV` (Anderson et al., 1999) is employed. The resulting eigenenergies and eigenstates are stored on disk. All data is stored as plain text files, so no additional software is required to read AlkCalc's output data, i.e., the eigenenergies and the eigenstates. It should be emphasised that all FORTRAN routines mentioned in this section are built into AlkCalc directly.

Stage II. This stage consists of the *library functions*. These functions read and process the stored eigenenergies and eigenstates. For the library functions, AlkCalc includes an optional Python interface, provided as the Python package pyalkcalc. Note that pyalkcalc is optional and not required to run AlkCalc.

Design choices. In contrast to shooting methods (Johnson, 2007), the B-spline approach yields all eigenstates within a chosen energy range from a single diagonalisation, without the need for initial guesses of the eigenenergies. This provides the full set of eigenstates required for the non-perturbative treatment of external fields. The compact support of the B-splines renders the matrices banded, which allows AlkCalc to use the banded LAPACK solver `DSBGVX` and reduces both memory requirements and computational cost compared to dense solvers. Since the diagonalisation is the computationally expensive step, AlkCalc computes the eigenenergies and eigenstates once and stores them on disk, such that derived quantities can be evaluated repeatedly without solving the eigenvalue problem again. The choice to store all data as plain text keeps the output portable and readable without adding software requirements, at the cost of larger files. However, the data requirements remain moderate: the stored eigenstates of a whole atom or ion species typically occupy between 10 and 100 MiB and stay below 1 GiB even for large bases (e.g., all states with principal quantum numbers below 120).

Research impact statement

The software AlkCalc has already been used in the publications (Euchner, Svendsen, et al., 2026; Euchner, Li, et al., 2026; Martins et al., 2026). In (Martins et al., 2026), AlkCalc was used to compute lifetimes of Rydberg states in $^{40}Ca^{+}$. In (Euchner, Svendsen, et al., 2026), AlkCalc was used to compute Rydberg-state lifetimes, eigenenergies, and transition matrix elements (and from these results Rydberg-state polarisabilities) for $^{88}Sr^{+}$. In (Euchner, Li, et al., 2026), AlkCalc was used to compute Rydberg-state lifetimes, eigenenergies, and transition matrix elements in the ions $^{40}Ca^{+}$ and $^{88}Sr^{+}$ and the atom ^{85}Rb. AlkCalc is actively maintained, and its development will be continued within further PhD work on Rydberg ions.

Planned work and future development

A planned extension of AlkCalc is the incorporation of relativistic corrections, which are essential for the precise treatment of heavy atoms and ions (Friedrich, 2017). The B-spline method employed by AlkCalc makes this extension straightforward, as it provides direct access to higher-order derivatives of numerically represented wave functions, which are required for the relativistic correction to the kinetic energy.

AI usage disclosure

For the preparation of the manuscript, Anthropic's Claude (Opus 5.5) was used for spell checking, consistency checks, and language and grammar suggestions. Anthropic's Claude (Opus 5) helped to search for inconsistencies and bugs in the code and to write a test suite for AlkCalc. All AI-assisted work was carefully reviewed by S.E., who takes full responsibility for it. The code for the library AlkCalc itself was written by S.E. personally.

Acknowledgements

S.E. is grateful for fruitful discussions with J. W. P. Wilkinson and W. S. Martins. This work was supported by the European Union through the ERC grant OPEN-2QS (Grant No. 101164443) and the Deutsche Forschungsgemeinschaft (DFG) within the Research Unit FOR 5413 (Grant No. 465199066) and the Research Unit FOR 5522 (Grant No. 499180199). We also acknowledge funding through JST-DFG 2024: Japanese-German Joint Call for Proposals on 'Quantum Technologies' (Japan-JST-DFG-ASPIRE 2024) under JST Grant No. JPMJAP24C2 and DFG Grant No. 554561799.

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